

CSE6214 Software Engineering Fundamentals

Tutorial 5

Part A: Discussion

Topic (Lecture 5): Design Concepts

1. Do you design software when you “write” a program? What makes software design different from coding?
2. How do we assess the quality of a software design?
3. What benefits can be obtained from designing based on the following concepts:
 - Abstraction
 - Software Architecture
 - Separation of Concerns
 - Modularity
 - Information Hiding
 - Functional Independence
 - Refinement
4. Discuss the relationship between the concept of information hiding as an attribute of effective modularity and the concept of module independence.
5. Does “refactoring” mean that you modify the entire design iteratively? If not, what does it mean?

Part B: Project

6. For each use case assigned to you, elaborate the process into a sequence of interaction between the user and the system, for example (make appointment use case):
 - a. Student enters date, time and lecturer name to make appointment.
 - b. System creates new appointment record in Appointment table.
 - c. System displays confirmation message to the student.
7. Based on the interactions, use Visual Paradigm to draw a sequence diagram for the use case. Refer to the following webpage on how to draw sequence diagrams:

http://www.visual-paradigm.com/support/documents/vpuserguide/94/2577/7025_creatingsequ.html